

Flow Matching · arXiv 2210.02747

Visionary

11+3 · seat 1 = Kanta

FM→BEHAVIOR map · 6 follow-ups
path design for action statistics

Sep 23 paper club · VLA Architectures

Spine: OT-CFM objective + path design for action statistics

Lipman et al. · Meta AI / Weizmann · pick a path, regress the VF

Visionary arc · one slide



Internalize from THIS paper

OpenPI loss = CFM · OT target $\approx a - \varepsilon$ along $a_t = (1-t)\varepsilon + t a$
Straight paths $\Rightarrow$ fewer NFEs (~ 10 Euler) · Diffusion $\subset$ FM — “back to DDPM” needs a reason

Don't overclaim: DP $\neq$ FM · correlated noise / N=15 $\neq$ Lipman · IN FID $\neq$ robot KPI

Internalize · four facts from THIS paper

1

OpenPI loss = CFM

Not score matching.
Conditional VF regression
on action chunks.

2

OT target $\approx a - \epsilon$

along interpolant
 $a_t = (1-t)\epsilon + t a$
constant direction.

3

Straight $\Rightarrow$ fewer NFEs

Permission for ~ 10 Euler
(Table 1 / Fig 7).
Latency = robot Hz.

4

Diffusion $\subset$ FM

VE/VP are special cases.
“Back to DDPM” needs
a reason — not default.

New apps (don't confuse with paper claims)

contact-rich soft schedules

multi-sample test-time selection

FAST+FM gated by likelihood

sim-to-real as second path

Conditional Flow Matching · the objective card

Thm 2: $\nabla \text{CFM} = \nabla \text{FM}$ · never need the marginal VF

Intractable FM

$$L_{FM} = E_{t, p_t} \|v_t(x) - u_t(x)\|^2$$

needs marginal generating field u_t

hard — mix over all data

Tractable CFM ★

$$L_{CFM} = E_{t, q, x} \|v_t - u_t(x|x_1)\|^2$$

per-sample conditional field

same gradients (Thm 2)

OT-CFM recipe (OpenPI / π_0 action expert)

1. Sample path

$$\begin{aligned} t &\sim U(0,1) \\ \varepsilon &\sim N(0,I), \quad a \sim \text{data} \\ a_t &= (1-t)\varepsilon + t a \end{aligned}$$

2. Target VF

$$\begin{aligned} u &\approx a - \varepsilon \\ &\text{(constant direction)} \\ \sigma_{\min} &\text{softens endpoint} \end{aligned}$$

3. Regress

$$\begin{aligned} &\|v_{\theta}(a_t, t | o) - u\|^2 \\ &\text{condition on obs/VLM} \\ &\rightarrow \text{action-chunk expert} \end{aligned}$$

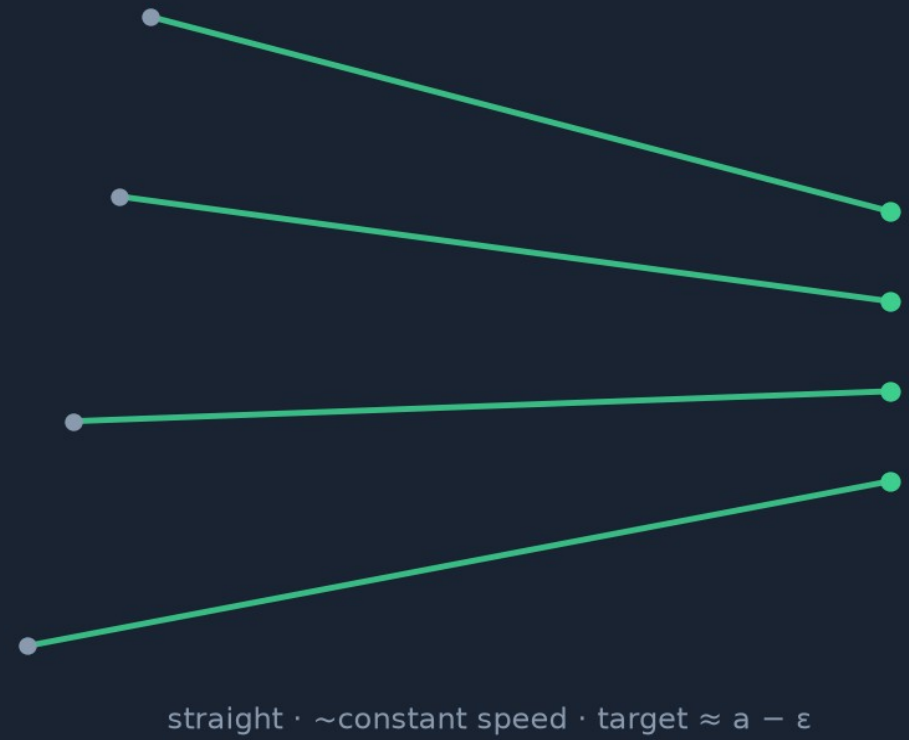
OT straight paths vs diffusion curved paths

Paper Figs 2-3 · constant-direction VF · fewer NFEs for same error

Diffusion / VP path



OT displacement path



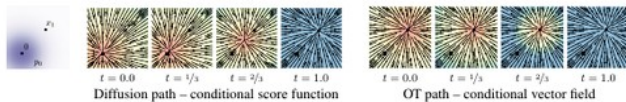


Figure 2: Compared to the diffusion path’s conditional score function, the OT path’s conditional vector field has constant direction in time and is arguably simpler to fit with a parametric model. Note the blue color denotes larger magnitude while red color denotes smaller magnitude.

which, in contrast to the diffusion conditional VF (equation 19), is defined for all $t \in [0, 1]$. The conditional flow that corresponds to $u_t(x|x_1)$ is

$$\psi_t(x) = (1 - (1 - \sigma_{\min})t)x + tx_1, \quad (22)$$

and in this case, the CFM loss (see equations 9, 14) takes the form:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,q(x_1),p(x_0)} \left\| v_t(\psi_t(x_0)) - \left(x_1 - (1 - \sigma_{\min})x_0 \right) \right\|^2. \quad (23)$$

Allowing the mean and std to change linearly not only leads to simple and intuitive paths, but it is actually also optimal in the following sense. The conditional flow $\psi_t(x)$ is in fact the Optimal Transport (OT) *displacement map* between the two Gaussians $p_0(x|x_1)$ and $p_1(x|x_1)$. The OT *interpolant*, which is a probability path, is defined to be (see Definition 1.1 in McCann (1997)):

$$p_t = [(1-t)\text{id} + t\psi]_* p_0 \quad (24)$$

where $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the OT map pushing p_0 to p_1 , id denotes the identity map, i.e., $\text{id}(x) = x$, and $(1-t)\text{id} + t\psi$ is called the OT displacement map. Example 1.7 in McCann (1997) shows, that in our case of two Gaussians where the first is a standard one, the OT displacement map takes the form of equation 22.

Intuitively, particles under the OT displacement map always move in straight line trajectories and with constant speed. Figure 3 depicts sampling paths for the diffusion and OT conditional VFs. Interestingly, we find that sampling trajectory from diffusion paths can “overshoot” the final sample, resulting in unnecessary backtracking, whilst the OT paths are guaranteed to stay straight.

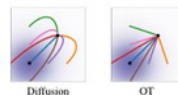


Figure 3: Diffusion and OT trajectories.

Figure 2 compares the diffusion conditional score function (the regression target in a typical diffusion methods), i.e., $\nabla \log p_t(x|x_1)$ with p_t defined as in equation 18, with the OT conditional VF (equation 21). The start (p_0) and end (p_1) Gaussians are identical in both examples. An interesting observation is that the OT VF has a constant direction in time, which arguably leads to a simpler regression task. This property can also be verified directly from equation 21 as the VF can be written in the form $u_t(x|x_1) = g(t)h(x|x_1)$. Figure 8 in the Appendix shows a visualization of the Diffusion VF. Lastly, we note that although the conditional flow is optimal, this by no means imply that the marginal VF is an optimal transport solution. Nevertheless, we expect the marginal vector field to remain relatively simple.

5 RELATED WORK

Continuous Normalizing Flows were introduced in (Chen et al., 2018) as a continuous-time version of Normalizing Flows (see e.g., Kobayev et al. (2020); Papamakarios et al. (2021) for an overview). Originally, CNFs are trained with the maximum likelihood objective, but this involves expensive ODE simulations for the forward and backward propagation, resulting in high time complexity due to the sequential nature of ODE simulations. Although some works demonstrated the capability of CNF generative models for image synthesis (Grathwohl et al., 2018), scaling up to very high dimensional images is inherently difficult. A number of works attempted to regularize the ODE to be easier to solve, e.g., using augmentation (Dupont et al., 2019), adding regularization terms (Yang & Karniadakis, 2019; Finlay et al., 2020; Onken et al., 2021; Tong et al., 2020; Kelly et al., 2020), or stochastically sampling the integration interval (Du et al., 2022). These works merely aim to regularize the ODE but do not change the fundamental training algorithm.

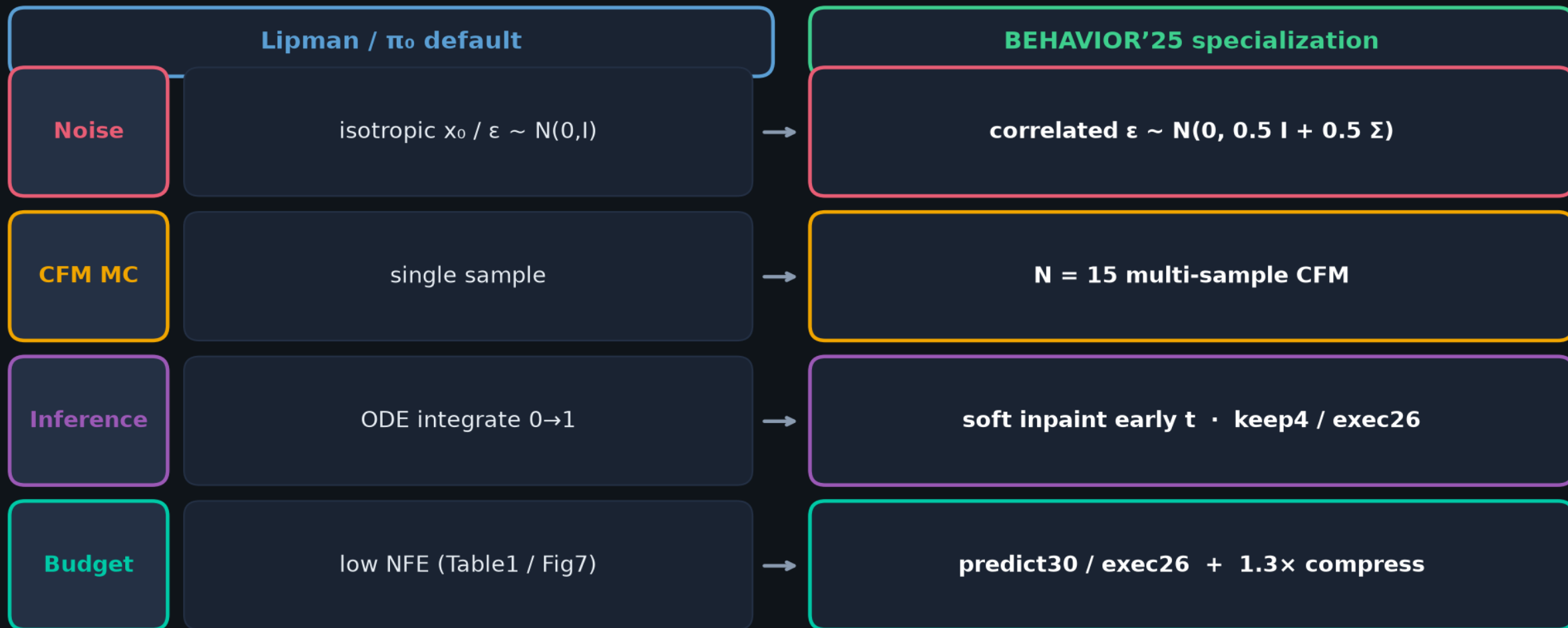


THE map

FM → BEHAVIOR specialization

FM → BEHAVIOR map · what changed for robot stats

Lipman'22 foundation → 2025 BEHAVIOR winners on Pi0.5 / OpenPI





THE talk figure

Six follow-ups for 2026

6 follow-ups to champion · 2026 BEHAVIOR / OpenPI

1

SPD Σ_t theory

CFM with full SPD Σ_t
closed-form u · publishable

2

OT vs VP vs DDPM

ablation inside OpenPI
freeze VLM · swap path only

3

Adaptive NFE

early-exit flow when
 $\|v_\theta\|$ small / confidence

4

π_0 curvature / reflow

measure trajectory curvature
try reflow if curved

5

Action-chunk NLL

change-of-variables
as health metric

6

Per-embodiment Σ

Σ paths in one FM expert
across robots / hands

New apps · after the paper

Contact-rich soft schedules

Path schedules that
respect contact modes
/ SE(3) constraints

Multi-sample test-time

$N > 1$ CFM samples →
select by likelihood
or critic

FAST + FM gated

Token FAST when sure;
FM when multimodal;
gate by NLL

Sim→real second path

Domain gap as a
second probability
path to regress

Still true: next wins = path design for action statistics

Not another ImageNet U-Net bake-off. Measure NFE-success Pareto under identical VLA encoders.

Don't overclaim

DP $\neq$ FM

Diffusion Policy = action-space DDPM. Conceptual bridge only. Likely no Lipman cite.

Σ / N=15 $\neq$ Lipman

Correlated noise + multi-sample CFM are BEHAVIOR specializations, not paper defaults.

IN FID $\neq$ robot KPI

ImageNet tables don't transfer to success rate / Hz / contact richness.

Conditional OT $\neq$ marginal OT

Authors admit marginal VF need not be W_2 -optimal — branding can overclaim.

Concurrent literature

Rectified Flow / interpolants overlap — cite them; claim framing+scale+OT emphasis.

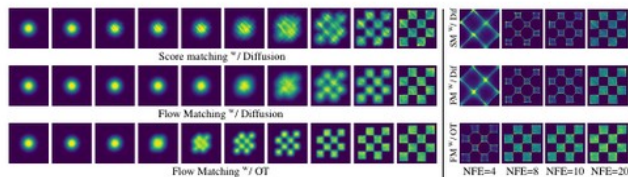


Figure 4: (left) Trajectories of CNFs trained with different objectives on 2D checkerboard data. The OT path introduces the checkerboard pattern much earlier, while FM results in more stable training. (right) FM with OT results in more efficient sampling, solved using the midpoint scheme.

In order to speed up CNF training, some works have developed simulation-free CNF training frameworks by explicitly designing the target probability path and the dynamics. For instance, Rozen et al. (2021) consider a linear interpolation between the prior and the target density but involves integrals that were difficult to estimate in high dimensions, while Ben-Hamu et al. (2022) consider general probability paths similar to this work but suffers from biased gradients in the stochastic minibatch regime. In contrast, the Flow Matching framework allows simulation-free training with unbiased gradients and readily scales to very high dimensions.

Another approach to simulation-free training relies on the construction of a diffusion process to indirectly define the target probability path (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song & Ermon, 2019). Song et al. (2020b) shows that diffusion models are trained using denoising score matching (Vincent, 2011), a conditional objective that provides unbiased gradients with respect to the score matching objective. Conditional Flow Matching draws inspiration from this result, but generalizes to matching vector fields directly. Due to the ease of scalability, diffusion models have received increased attention, producing a variety of improvements such as loss-rescaling (Song et al., 2021), adding classifier guidance along with architectural improvements (Dhariwal & Nichol, 2021), and learning the noise schedule (Nichol & Dhariwal, 2021; Kingma et al., 2021). However, (Nichol & Dhariwal, 2021) and (Kingma et al., 2021) only consider a restricted setting of Gaussian conditional paths defined by simple diffusion processes with a single parameter—in particular, it does not include our conditional OT path. In another line of works, (De Bortoli et al., 2021; Wang et al., 2021; Peluchetti, 2021) proposed finite time diffusion constructions via diffusion bridges theory resolving the approximation error incurred by infinite time denoising constructions. While existing works make use of a connection between diffusion processes and continuous normalizing flows with the same probability path (Maoutsa et al., 2020b; Song et al., 2020b; 2021), our work allows us to generalize beyond the class of probability paths modeled by simple diffusion. With our work, it is possible to completely sidestep the diffusion process construction and reason directly with probability paths, while still retaining efficient training and log-likelihood evaluations. Lastly, concurrently to our work (Liu et al., 2022; Albergo & Vanden-Eijnden, 2022) arrived at similar conditional objectives for simulation-free training of CNFs, while Neklyudov et al. (2023) derived an implicit objective when u_t is assumed to be a gradient field.

6 EXPERIMENTS

We explore the empirical benefits of using Flow Matching on the image datasets of CIFAR-10 (Krizhevsky et al., 2009) and ImageNet at resolutions 32, 64, and 128 (Chrabaszcz et al., 2017; Deng et al., 2009). We also ablate the choice of diffusion path in Flow Matching, particularly between the standard variance preserving diffusion path and the optimal transport path. We discuss how sample generation is improved by directly parameterizing the generating vector field and using the Flow Matching objective. Lastly we show Flow Matching can also be used in the conditional generation setting. Unless otherwise specified, we evaluate likelihood and samples from the model using `dopri5` (Dormand & Prince, 1980) at absolute and relative tolerances of $1e-5$. Generated samples can be found in the Appendix, and all implementation details are in Appendix E.

Visionary · close / Q&A

Next wins = path design for action statistics.

Spine reminder:

FM = pick a path, regress the VF; OT → straighter/faster
→ DP chunks → π_0 OT-CFM head → BEHAVIOR fixes robot stats

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